Mahsa Raeisinezhad

Description

Highly skilled AI Research Scientist with 4+ years of experience designing, developing, and deploying ML solutions. Skilled in Python, C++, SQL, and parallel/distributed computing, with publications in leading journals.

Work Experience

Dec 2025-Present **Technical Lead, Robotics AI Research**, *FieldAI*

- Led an end-to-end platform that turns human motion-capture data into robot-ready skills, owning product, architecture, infrastructure and delivery, and cutting the time from capture to robot-ready motion by 80%
- Defined and owned the product requirements and delivery roadmap from production launch to robot deployment. Aligned three teams and cut the planned time to first customer delivery by two months.
- Led a cross-functional team across web, machine learning, infrastructure and customer engineering; set priorities, reviewed code and led design reviews.

Oct 2025-Oct 2026 **AI Engineer**, *Apple*

- Automating internal workflows and replacing manual processes with intelligent AI-driven systems.
- Built an internal library that automated RAG-based projects across the organization, selecting the best-fit RAG approach for new use cases and cutting time from problem to working solution by 40%.
- Built an automated platform that let anyone in the organization, including non-engineers such as leadership, ship vibe-coded apps to production in under 20 minutes, replacing manual release steps with repeatable, tested rollouts.
- Trained engineers in applied AI, including LLMs, RAG and AI-assisted development, through workshops and hands-on guidance, helping teams adopt AI in their day-to-day work.

Jan 2024–April 2025 **Research Scientist**, *Lighthouse Regulatory Consulting Group*

- Built a Deep Reinforcement Learning solution with POMDPs for real-time sensor-driven decision making, reducing risk and uncertainty.
- Authored and documented an NIH grant proposal for the development of biomedical devices, including detailed research plans, technical specifications, and regulatory considerations.
- Contributed to the AI-driven predictive maintenance system for medical device manufacturing, boosting equipment reliability and production efficiency, resulting in a 20% increase in annual savings.
- Integrated advanced robotics and machine learning algorithms into existing control systems, enhancing production throughput by 40% and reducing energy consumption by 15%.

Aug 2023-Jan 2024 **Technical Lead, AI Engineering**, *Katz Graduate School of Business - UPitt*

- Developed deep learning pipelines for anomaly detection and predictive maintenance using IoT devices, boosting system efficiency by 30% and reducing system downtime by 25%.
- Led a five-person cross-functional engineering and analytics team.

Aug 2021-Jan 2024 **Ph.D Research Assistant**, *University of Pittsburgh*

- Designed an explainable (Kernel-SHAP + LIME) double-agent actor-critic Deep Reinforcement Learning-based control algorithm for replacing manual control strategies resulting in a 50% cost reduction in nuclear power plant operations and maintenance.

Education

- M.S., Mechanical Engineering, University of Pittsburgh 2021–2023
- M.S., Computer Science, Rowan University 2019–2021
- B.S., Computer Science, Shiraz University 2013–2017

Publications

Mahsa Raeisinezhad, et al. "Explainable, Deep Reinforcement Learning–Based Decision Making for Operations and Maintenance." NUCLEAR TECHNOLOGY

Mahsa Raeisinezhad, et al. "Design Optimization of a Pneumatic Soft Robotic Actuator Using Model-Based Optimization and Deep Reinforcement Learning" Frontiers in Robotics and AI

Mahsa Raeisinezhad, et al. "Intelligent Soft Robotic Pad for Pressure Injury Prevention" IEEE

Certificate

Artificial Intelligence on Microsoft Azure (Coursera)

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